INCH-POUND

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DETAIL SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M25A1 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M25A1 Cartridge Actuated Thruster Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550 Ammunition, General Specification For

MIL-T-50713 Thruster, Cartridge Actuated, M25A1 Remanufactured Metal Parts Assembly

MIL-T-60561 Thruster, Cartridge Actuated, M25A1 Metal Parts Assembly

Comments, suggestions, or questions on this document should be addressed to Department of the Navy, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

AMSC N/A FSC 1377

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 Ammunition Lot Numbering and Ammunition Data Card

MS28741 Hose Assembly, Detachable End Fitting, Medium Pressure

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10520589 Cartridge, Impulse, M127 Assembly

8797741 Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1

8797742 Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1

DL11730533 Thruster, Cartridge Actuated, M25A1 Assembly

9217579 Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1

U.S. Air Force

TPO 00-554-3103 Thruster, Cartridge Actuated, M25A1

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQ Z1.4 Sampling Procedures And Tables For Inspection By Attributes

ANSI/ISO/ASQ A8402 Quality Assurance Terms And Definitions

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(Copies of this document are available online at http://e-standards.asq.org/ or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Inspection, Radiographic

(Copies of these documents are available online at www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.2 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Components.

3.2.1 Primary components. For the purpose of this specification, the M127 impulse cartridge (Drawing 10520589) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of thrusters; however, one primary component production lot may be used in more than one thruster production lot.

3.2.2 Component parts. All component parts shall meet the applicable requirements of MIL-T-50713 or MIL-T-60561 prior to assembly.

3.3 Torque. Torque requirements shall be as specified in table I. No movement shall occur upon application of the breakaway torque.

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TABLE I. Torque requirements.

Components	A	B	C
	Run On, min (inch pounds)	Seating or bottoming (including A) (inch pounds)	Breakaway, min (inch pounds)
Head and retainer guide	50	300 ± 20	200
Retainer guide and sleeve after insertion of cartridge	—	300 ± 20	200
Head and body[1/]	50	600 ± 20	400
Cap screws	—	20 minimum	10

[1/] At no time will torque be applied between the piston and body assembly.

3.4 Trunnion rotation. Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.5 Residual magnetism. The thruster shall not deflect the indicator of the compass more than five degrees in either direction (see 4.3.2).

3.6 Radiographic examination. Each thruster assembly shall be radiographically examined for defects (see 4.3.3).

3.7 Functional.

3.7.1 Load. The thruster fired under load (see 6.3) shall perform as follows:

3.7.1.1 Thrust. With the piston connected to a 50 pound weight, the thruster shall deliver sufficient thrust to overcome a 3,000 pound resistive load, horizontally, throughout its total stroke of 5 inches minimum. The resistive load shall not be applied on the piston before initial movement.

3.7.1.2 Operating time. The time from initial thrust rise to completion of the piston stroke shall be 30 milliseconds, minimum, and 90 milliseconds, maximum.

3.7.1.3 Maximum thrust. A thrust-time record shall be made for each ballistic firing. The maximum thrust will be computed from that trace and recorded for information purposes on the firing sheet.

3.7.2 Ignition delay. The ignition delay time shall not exceed 65 milliseconds (see 6.5.1).

3.7.3 Temperature. The thruster shall comply with all the ballistic requirements throughout the temperature range of -65 to +200°F.

3.7.4 Initiation. The thruster shall be fired by a source delivering an output pressure of 700 + 25, - 0 pounds per square inch gage (psig) at the end of 15 feet of aircraft hose, MS28741-4 at 70 ± 5°F.

3.7.5 Misfires. There shall be no misfires.

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3.7.6 Mechanical failures. The thruster shall function without mechanical failure (see 6.5.2).

3.8 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections.

a. Quality conformance inspection

4.1.1 Quality assurance terms and definitions. Reference shall be made to ANSI/ISO/ASQ A8402 to define the quality assurance terms used.

4.1.2 Inspections. Inspections shall be in accordance with MIL-A-2550 and ASTM E 1742.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation, and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

4.2.1.2 Size of lot. The production lot size shall be as specified in the contract (see 6.2).

4.2.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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4.2.2.1 Cartridge, Drawing 10520589, covering a detail of Drawing 11730533.

Characteristics	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	Marking missing, incorrect, or illegible	Visual
Major		
101	Foreign matter present	Visual
Minor		
201	Case scratched or dented	Visual

4.2.2.2 Retainer, Guide Subassembly, consisting of Guide Retainer and Pellet Assembly 10521616, Shear Pin 10521618, Firing Pin 8593864, and O-Ring MS29513-12, covering a detail of Drawing 11730533.

Critical		
1	Striker hole obstructed	Visual
Major		
101	Foreign matter	Visual
Minor		
None defined		

4.2.2.3 Head Sub-Assembly, consisting of Head 10521455, Guide Retainer and Pellet Assembly 10521616, Sleeve 10521617, Shear Pin 10521618, Firing Pin 8593864, Cartridge 10520589, and O-Rings MS29513-12 and MS29513-212, covering a detail of Drawing 11730533.

Critical		
None defined		
Major		
101	Thread damaged or not full	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	Visual
Minor		
201	Excess sealant not removed	Visual

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4.2.2.4 Assembly, Drawing 11730533, less NAS845 and NAS814.

Critical		
1	Foreign matter in gas port	Visual
2	Shear pin not engaged in firing pin	Gage
Major		
101	Total length	Gage
102	Thread damaged	Visual
Minor		
201	Sealant or coating compound missing	Visual

4.2.2.5 Assembly, Drawing 11730533.

Critical		
1	Improper assembly	Visual/X-ray
2	Any part missing	Visual/X-ray
3	Metal defective	Visual
Major		
101	Marking missing, incorrect, or illegible	Visual
102	Foreign matter	Visual
Minor		
201	Trunnion not free to rotate 360 degrees	Manual
202	Location of labels incorrect	Visual
203	Protective cap or plug missing	Visual
204	Total length	Visual

4.2.2.6 Box, Fiberboard (Inner), Ammunition, Drawing 8797741.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

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4.2.2.7 Box, Ammunition, Packing, Drawing 8797742.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.2.2.8 Box, Fiberboard (Outer), Alternative, Ammunition, Packing, Drawing 9217579.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.

4.2.3.1.1 Retainer guide and sleeve. Each retainer guide and sleeve subassembly shall be subjected to a breakaway torque. Failure of any subassembly to comply with the requirements of column C, table I in 3.3 shall be cause for rejection of the subassembly.

4.2.3.1.2 Head and retainer guide. Each head and retainer guide subassembly shall be subjected to a breakaway torque. Failure of any subassembly to comply with the requirements of column C, table I in 3.3 shall be cause for rejection of the subassembly.

4.2.3.1.3 Head and body. Each head and body assembly shall be subjected to a breakaway torque. Failure of any assembly to comply with the requirements of column C, table I in 3.3 shall be cause for rejection of the assembly.

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4.2.3.2 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any unit to comply with 3.5 shall be cause for rejection of the assembly.

4.2.3.3 Functional tests. The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.7. The units shall be temperature conditioned as provided in table II and 4.3.4.2. Failure of any unit to comply with 3.7.1 through 3.7.6 shall be cause for rejection of the lot.

TABLE II. Conditioning temperature.

Sample size	Method	Temperature (°F)
7	Load (see 3.7.1)	-65 to -70
7	Load (see 3.7.1)	65 to 75
7	Load (see 3.7.1)	195 to 205

Ballistic test samples for small lots shall be:

Lot size	Ballistic Sample
Less than 9	4
9 – 25	6
26 – 50	8
51 – 90	10
91 – 150	14
Over 150	21

4.2.3.3.2 Retest. There shall be no retests.

4.2.3.3.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test thruster, the original test results shall be discarded and that test reconducted.

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench, set at the torque requirements specified in table I, shall be used in conjunction with a fixture to determine the adequacy of the joints.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in the north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

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4.3.3 Radiographic examination. All assembled thrusters shall be radiographically examined in accordance with ASTM E 1742, with the exception that an thruster assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The thrusters shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All thrusters shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The thrusters shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective thrusters found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Cartridge, Impulse, P/N 10520589
b. Firing Pin, P/N 8593864
c. Head, P/N 10521455
d. Shear Pin, P/N 10521618
e. Trunnion, P/N 8594482
f. Guide, Retainer, P/N 10521616
g. Sleeve, P/N 10521617
h. Body, P/N 10534179
i. Keys (3), P/N 10534180
j. Piston, P/N 10534181
k. Piston, unlocking, P/N 10534182
l. Ring, P/N 10534183
m. Spring, P/N 10534184
n. O-ring, P/N MS29513-12
o. O-ring, P/N MS29513-113
p. O-ring (2), P/N MS29513-212
q. O-ring, P/N MS29513-214
r. Cap Screw (2), P/N MS16998-9

4.3.4 Functional.

4.3.4.1 Ballistics. The test sample shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures.

4.3.4.2 Temperature conditioning. Each assembled thruster shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning thrusters at -65°F, all condensation shall be removed from the exterior surfaces before returning the thruster to the conditioning chamber.

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4.4 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification

b. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2)

c. Production test activity and lot size (see 4.1 and 4.2.1)

d. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.3.3)

e. Level of packing and marking (see 5.1, 6.5)

f. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of ``Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Load firing. The expected performance of the thruster, under conditions simulating those existing in aircraft, is obtained when the thruster satisfactorily meets the conditions of the "load" firing.

6.4 Hazard notice. The thruster described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should

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recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.5 Packaging

6.5.1 Packing and marking. Packing and marking should be Level A or B, as specified by the contracting activity (see 6.2). Level A should be in accordance with Drawings 8797741 and 8797742. Level B should be in accordance with Drawings 8797741 and 9217579.

6.5.1.1 Lot numbers. Lot numbers should be in accordance with MIL-STD-1168.

6.5.2 Air Force packaging. Air Force packaging should be in accordance with TPO 00-554-3103.

6.6 Definitions.

6.6.1 Ignition delay. Ignition delay is the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure-time record.

6.6.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. The changes in this issue were authorized by ECP 09E215A038DB.

Custodian:
Navy - OS
Air Force - 99

Preparing activity
Navy - OS
(Project 1377-2010-004)

Review activity:
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